



AI-BASED INTERACTIVE SYSTEM IN HUMAN-COMPUTER INTERACTION USING NATURAL LANGUAGE PROCESSING FOR DISABLED PEOPLE

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Abstract

This project develops an offline voice-controlled desktop automation system that enables users to operate a computer using spoken commands. The system integrates Natural Language Processing (NLP) and Automatic Speech Recognition (ASR) technologies using Vosk, PocketSphinx, and CMUSphinx. It supports functions such as mouse control, scrolling, browser automation, application launching, camera control, and volume management. By processing commands locally without internet connectivity, the system provides improved privacy, reliability, and accessibility, particularly for users with physical disabilities.

Problem Statement

Traditional computer interaction relies on keyboards and mouse devices, which can be difficult for users with physical disabilities. Additionally, many voice-recognition systems depend on internet connectivity, raising privacy and reliability concerns. Therefore, an offline voice-controlled desktop automation system is needed to provide accessible, hands-free computer interaction.

Objective

1 Develop an offline voice-controlled desktop automation system using Vosk, PocketSphinx, and CMUSphinx speech recognition technologies.

2 Implement voice-command functionalities for:

- Mouse control
- Scrolling
- Browser automation
- Application launching
- Multimedia and volume control

3 Evaluate system accuracy, reliability, and usability through performance testing and Word Error Rate (WER) analysis.

4 Analyze the effect of environmental noise on speech recognition performance.

Methodology

Development Approach

Prototype Development Methodology

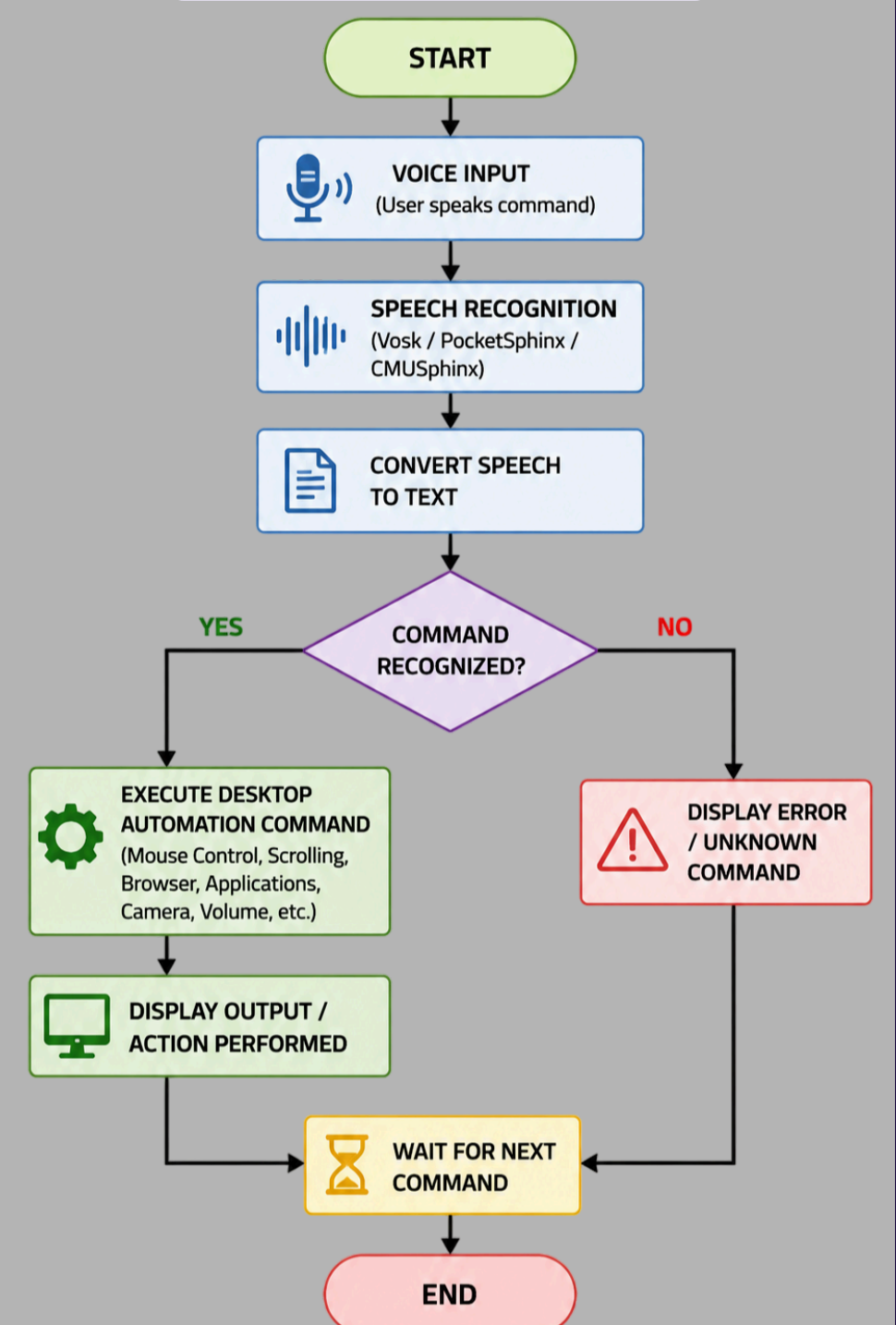
Tools & Technologies

- Python
- Vosk
- PocketSphinx
- CMUSphinx
- PyAutoGUI
- Webbrowser Module
- Subprocess Module
- SoundDevice

Development Stages

- Requirement Analysis
- System Design
- Speech Recognition Integration
- Command Processing
- Desktop Automation Implementation
- Testing & Evaluation
- Performance Analysis

Flowchart



Outcome

Functionalities Successfully Implemented

- ✓ Mouse Movement
- ✓ Left Click / Right Click / Double Click
- ✓ Scrolling
- ✓ Browser Control
- ✓ YouTube Access
- ✓ Calculator Launching
- ✓ Camera Control
- ✓ Volume Control

Key Findings

- Vosk achieved the highest recognition accuracy.
- Offline speech recognition provides improved privacy.
- Real-time command execution was successfully achieved.
- The system remained functional without internet connectivity.
- The solution improves accessibility for users with physical disabilities.

Conclusion

The proposed offline voice-controlled desktop automation system enables hands-free computer interaction through speech commands, providing an accessible, reliable, and privacy-preserving solution for users with physical disabilities.